Agentic Retrieval-Augmented Generation (RAG) System for Explainable Legal Case Analysis

Milestone -1

Course : GEN-AI and LLM
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Domain Note

1. Problem Statement

In the field of law, there is a vast amount of cases and decisions that need to be studied to gather pertinent information. However, there are no available tools that help retrieve data by semantic search or relevance. Consequently, users have to read numerous papers until they get some useful results.

2. Who It Affects and Why It Matters

This problem affects:

- **Lawyers**, who need quick access to relevant case precedents
- **Law students**, who must understand legal reasoning
- **Researchers**, who analyze judicial trends

Inefficient retrieval and interpretation of legal information can lead to missed insights and reduced productivity.

3. Limitations of Existing Solutions

Current systems suffer from:

- Keyword-based retrieval with poor semantic understanding
- Lack of explanation or summarization
- High manual effort to interpret results

These limitations highlight the need for intelligent systems that can both retrieve and explain legal content.

4. Proposed Approach

We propose a **Retrieval-Augmented Generation (RAG) system** with an agent-based design. The system:

- Uses **embeddings and FAISS** for semantic retrieval
- Uses **Large Language Models (LLMs)** to generate explanations
- Incorporates modular **agents** (retriever, analyzer, explainer)

Given a natural language query, the system retrieves relevant documents and generates a clear, context-aware response with reasoning.

5. Justification of Approach

The proposed system:

- Improves retrieval using semantic search
- Reduces hallucination through RAG
- Enhances usability via explainable outputs
- Provides a scalable and modular architecture

6. Current Progress and Next Steps

For Milestone 1, we have:

- Built a working data pipeline
- Implemented semantic retrieval (FAISS + embeddings)
- Developed a basic RAG system for answer generation

Future work includes integrating real legal datasets and improving evaluation.

Project Pipeline – Agentic Legal RAG System

1. Input Query

The system begins by accepting a natural language legal query from the user (e.g., *"Explain fraud-related legal cases"*). This serves as the entry point for the entire pipeline

2. Query Expansion

To improve retrieval quality, the input query is expanded into multiple variations using predefined templates:

- Original query
- Legal explanation variant
- Case law-related variant
- Judgment reasoning variant

This step ensures broader semantic coverage and improves document retrieval accuracy.

3. Retrieval Agent (Vector Search)

Each expanded query is passed to the **retrieval agent**, which:

- Uses SentenceTransformers to generate embeddings
- Searches a FAISS vector index
- Retrieves top-k semantically relevant documents

All retrieved documents are aggregated and deduplicated.

4. Decision Agent (Control Logic)

A lightweight decision agent evaluates whether sufficient context has been retrieved:

- If documents are insufficient → triggers query refinement
- Performs a second retrieval step with a more detailed query

This introduces **multi-step reasoning and adaptive search behavior**.

5. Analyzer Agent (Context Builder)

The analyzer agent processes the retrieved documents by:

- Filtering and selecting top relevant results
- Constructing a coherent context for reasoning

This ensures that only meaningful information is passed to the language model

6. LLM Reasoning (Answer Generation)

The processed context and original query are passed to an LLM (GPT-4o-mini):

- Generates a structured legal explanation
- Avoids hallucination using prompt constraints
- Produces reasoning-based responses

7. Explainability Layer

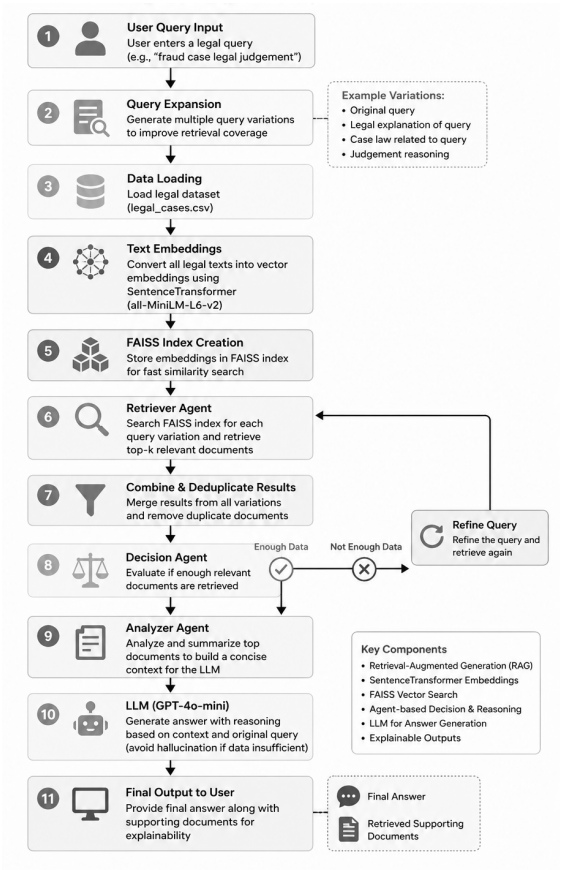
The system improves transparency by:

- Displaying retrieved source documents
- Providing reasoning-based answers
- Allowing users to trace outputs back to context

8. User Interface (Frontend)

A Streamlit-based UI enables:

- User query input
- Display of retrieved sources
- Final AI-generated answer



Results

Prompt -

```
You are a legal assistant.

Use ONLY the information from the retrieved documents.

If the information is insufficient, say:
"Not enough data available."
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Output-

```
🔍 Retrieved Sources:

Source 1: Election Fraud - 20004 Fraud is not a word that one uses lightly. Indeed, in the context of a Presidential ele
ction, using it exposes one to dismissal as the very thought of it is distasteful. This ti...

Source 2: Justice Dept. Seeks  #36;2.5 Bln from Rigas (Reuters) Reuters - The Justice Department has asked a\judge to en
ter a  #36;2.53 billion judgment against two Rigas\family members, citing the amount it cl...

Source 3: Galvin charges firm with fraud for failing to admit guilt It's not enough to pay a fine for breaking securitie
s laws in Massachusetts. State officials want a public admission of guilt, too....


🧠 Final AI Answer:

Based on the retrieved documents, there are several instances of fraud-related legal cases that illustrate different asp
ects of fraud in various contexts.

1. **Election Fraud**: The document mentions that there is clear and unmistakable evidence of fraud in the context of a
Presidential election. This suggests that there are serious allegations being made regarding the integrity of the electo
ral process. The use of the term "fraud" in this context indicates that the evidence presented is substantial enough to
warrant legal scrutiny, despite the general reluctance to use such a term lightly in political contexts.

2. **Adelphia Communications Corp. Case**: The Justice Department is seeking a judgment of approximately $2.53 billion a
gainst members of the Rigas family, founders of Adelphia Communications Corp. This case highlights a significant legal a
ction where the government claims that the Rigas family obtained money illegally. This indicates a serious allegation of
 financial fraud, likely involving securities or corporate governance violations.

3. **Galvin Charges Firm with Fraud**: In Massachusetts, state officials are pursuing a firm for fraud related to breaki
ng securities laws. The emphasis here is not only on the financial penalty but also on the demand for a public admission
 of guilt. This reflects a legal approach where accountability is sought not just through fines but also through public
acknowledgment of wrongdoing, which can serve to restore public trust in the regulatory framework.

4. **Internet Fraud**: The document discusses internet fraud, specifically mentioning "phishers" who use sophisticated m
ethods to deceive individuals into revealing their banking details. This type of fraud highlights the evolving nature of
 fraudulent activities in the digital age, where technology is exploited to commit crimes against unsuspecting victims.

In summary, these cases illustrate various dimensions of fraud, including electoral fraud, corporate fraud, regulatory c
ompliance failures, and internet fraud. Each case reflects the legal system's response to different types of fraudulent
behavior, emphasizing accountability, the pursuit of justice, and the protection of public trust.
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Prompt -

⚖️ Agentic Legal RAG System

Explainable Legal Case Analysis using AI

🔍 Enter your legal query

What are fraud-related legal cases?

Run Analysis

Example queries:

- What are fraud-related legal cases?
- Explain judgement reasoning in fraud cases
- What types of legal disputes occur in financial fraud?

Output -

🔍 Retrieved Sources

Source 1

quot;Phishing quot; lures German banking clients Two clients of Germany #39;s Postbank have fallen for an e-mail fraud that led them to reveal money transfer codes to a bogus Web site — the first case of this scam in German, prosecutors say.

Source 2

Court throws out apartheid cases A New York judge throws out legal cases against more than 30 firms, accused of having aided the apartheid-era South African Government.

Source 3

ING Reveals Improper Trading Case Details NEW YORK (Reuters) - A U.S. unit of Dutch financial services group ING Groep NV said on Wednesday a review of improper trading in its mutual funds showed only isolated incidents of impropriety.

🧠 Final AI Answer

Fraud-related legal cases typically involve allegations of deceptive practices intended to secure unfair or unlawful gain. These cases can encompass a variety of fraudulent activities, including but not limited to:

1. **Phishing Scams**: For example, the case involving clients of Germany's Postbank who were tricked into revealing money transfer codes through a fraudulent email. This represents a direct case of fraud where individuals are misled into providing sensitive information.
2. **Improper Trading**: The case involving ING Groep NV highlights issues of improper trading within mutual funds, where isolated incidents of misconduct can lead to legal scrutiny and potential penalties.
3. **Fake Trades**: The indictments against former natural-gas traders for reporting fake trades illustrate another form of fraud, where false information is disseminated to manipulate market perceptions or valuations.
4. **Antitrust Violations**: The upheld ruling against Visa and Mastercard for violating antitrust laws can also be seen as a form of fraud, where companies engage in practices that unfairly limit competition and harm consumers.
5. **Identity Theft**: The case of the world's largest ID theft felon, who pleaded guilty to a significant scam, represents a serious form of fraud that involves stealing personal information for financial gain.

These cases demonstrate various methods of fraud, each with legal implications that can lead to civil or criminal charges, depending on the nature and severity of the fraudulent activity.

Prompt -

🔍 Enter your legal query

What types of legal disputes occur in financial fraud?

Run Analysis

Example queries:

- What are fraud-related legal cases?
- Explain judgement reasoning in fraud cases

Output -

🔍 **Retrieved Sources**

Source 1

Legal Fight Menaces Aircraft Debt Ratings NEW YORK (Reuters) - Standard Poor's on Friday said ratings on about $30 billion of aircraft-backed debt could be threatened if creditors lost a legal dispute with bankrupt UAL Corp. <A HREF="http://www.investor.reuters.com/FullQuote.aspx?ticker=UALAQ.OB target=/stocks/quickinfo/fullquote">UALAQ.OB</A> unit United Airlines over the right to repossess aircraft.

Source 2

Jury Gets Enron, Merrill Lynch Case Jurors began deliberations in the trial of six former Enron Corp. and Merrill Lynch amp; Co. executives accused of fraud and conspiracy in the energy trader #39;s 1999 sale of three power-generating barges.

Source 3

Court throws out apartheid cases A New York judge throws out legal cases against more than 30 firms, accused of having aided the apartheid-era South African Government.

🧠 **Final AI Answer**

Not enough data available.

Evaluation of Retrieval Quality

To assess the performance of the system, basic retrieval evaluation was conducted:

- Relevance Check (Manual Evaluation):
  Retrieved documents were examined to determine their relevance to the query posed by the user.
- Top-k Retrieval Accuracy:
  The system returns top 3 documents (k = 3). In most test instances, there were 1–2 relevant documents retrieved by the system.

Conclusion

The current research has been successful in demonstrating an agentic legal RAG model that makes use of semantic search and AI reasoning for answering legal questions. In addition, the model searches through the case documents and provides explanations without generating hallucinations. This marks a critical step towards creating a more powerful legal AI assistant.